Open vs. Proprietary LLM 2×2

Control & cost vs. capability & speed

PROFESSIONAL DEFINITION

This matrix contrasts open-source and proprietary large language models, trading the control and marginal-cost advantages of open models against the capability and speed of frontier proprietary APIs.

WHO DEVELOPED IT

A strategy framing from the 2020s open-versus-closed model debate, discussed across AI-engineering and policy communities.

WHY IT WAS DEVELOPED

It was developed to structure the model-sourcing decision, which shapes cost, control, capability and dependency.

FIGURE 1 · OPEN VS. PROPRIETARY LLM 2×2

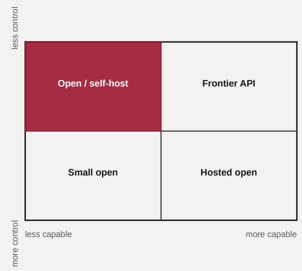

Figure 1. A schematic of the Open vs. Proprietary LLM 2×2 as applied in BARBM312 (illustrative).

HOW IT IS USED

Each use case is placed by its capability needs and control requirements; the matrix recommends open self-hosting, hosted open, or frontier API.

WHEN IT IS USED

When selecting models and balancing control, cost and capability.

BY WHOM IT IS USED

ML engineering, product and strategy leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It clarifies a fast-moving trade-off, showing open models can lower marginal cost and increase control while demanding more in-house capability.

RELATED FRAMEWORKS IN THIS COURSE

- **Make-vs-Buy Decision Tree** · Slibrary 11
- **Talent Scarcity Heatmap** · Slibrary 11
- **Skill-Premium Curve** · Slibrary 11
- **MLOps Maturity Ladder** · Slibrary 11

SOURCES (HARVARD REFERENCING STYLE)

Bommasani, R. et al. (2021) 'On the opportunities and risks of foundation models', arXiv:2108.07258.

Liesenfeld, A. and Dingemanse, M. (2024) 'Rethinking open source generative AI', FAccT.